ParkeSafe
FIND. BOOK. PARK SAFE.

ParkSafe

Smart Vehicle Parking & Online Slot Booking Application

FIND. BOOK. PARK SAFE.

TEAM MEMBERS:

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DEPARTMENT:

B.Voc (Bachelor of Vocation)

Software Development / Computer Applications

OVERVIEW

Introduction

ParkSafe is a smart parking application that helps people find and book parking spaces before they reach the parking location.

For Customers

- Find nearby parking locations easily.
- Check live available slots before arriving.
- Select Car, Bike, or EV parking.
- Choose custom date, time, and duration.
- Select exact floor and parking slot number.
- Receive a digital ticket with QR code instantly.

For Parking Admins

- Manage multiple parking locations.
- Configure multi-floor slot distribution.
- Scan customer QR codes for rapid check-in.
- Verify bookings and prevent location mismatch.
- Monitor real-time occupancy and daily revenue.
- Handle booking cancellations and refunds.

WORKFLOW

How ParkSafe Works

ParkSafe connects the customer, parking location, and admin in one unified digital system.

STEP 01

Search & Locate

Customer opens ParkSafe, selects city, and discovers nearby available parking locations in real time.

STEP 02

Select Slot

Chooses vehicle type (Car, Bike, EV), booking date, time duration, parking floor, and exact slot.

STEP 03

Book & Pay

Confirms reservation, completes fast payment via NETC FASTag or UPI, and gets instant digital ticket.

STEP 04

Scan & Park

Admin scans customer QR code at the gate for rapid check-in validation, parking, and smooth check-out.

KEY BENEFITS

Advantages of ParkSafe

Saves Time & Fuel

Customers book in advance and drive directly to their assigned slot without searching.

Real-Time Slot Visibility

Shows live available slots per floor and category (Car, Bike, and EV spots).

Multi-Floor Management

Organizes large parking complexes into easy-to-navigate floor and slot numbers.

Contactless QR Entry

Admin scans digital QR code for instant check-in and check-out at the gate.

Integrated Navigation

One-tap Google Maps directions directly guide drivers to the parking gate.

Complete Admin Control

Admins manage locations, slots, live check-ins, revenue trends, and refunds.

COMPARISON

Before and After ParkSafe

Before ParkSafe (Traditional Parking)

✕ Drivers search for parking spots manually on the road.

✕ No information about slot availability before arriving.

✕ Long wait times and manual paper ticket entry at gates.

✕ Multi-floor parking is confusing and difficult to navigate.

✕ Admins track vehicles and collections using manual registers.

✕ No proper mechanism for digital cancellations or refunds.

With ParkSafe (Smart Parking)

✓ Drivers search and book parking online in advance.

✓ Live slot availability visible per floor and vehicle type.

✓ Fast QR code scan for contactless check-in and check-out.

✓ Pre-allocated floor and slot numbers guide drivers clearly.

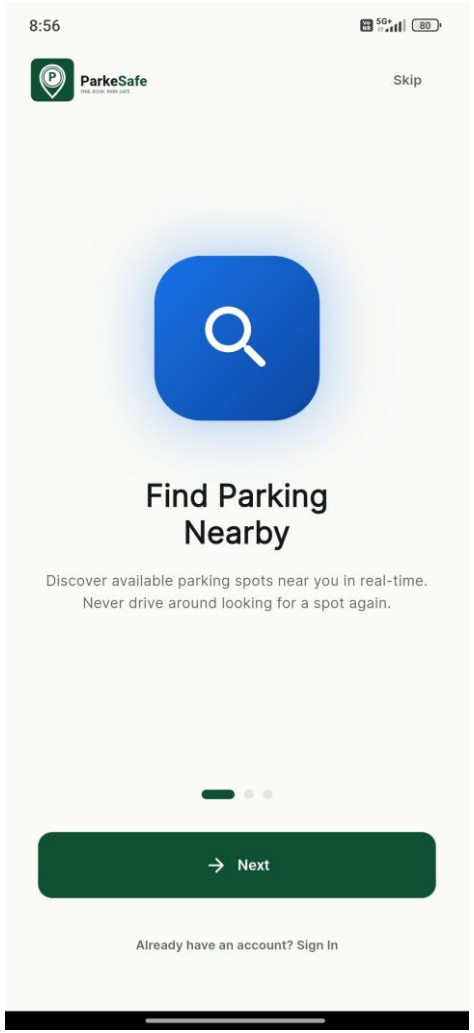
✓ Admin Panel manages live occupancy, slots, and revenue.

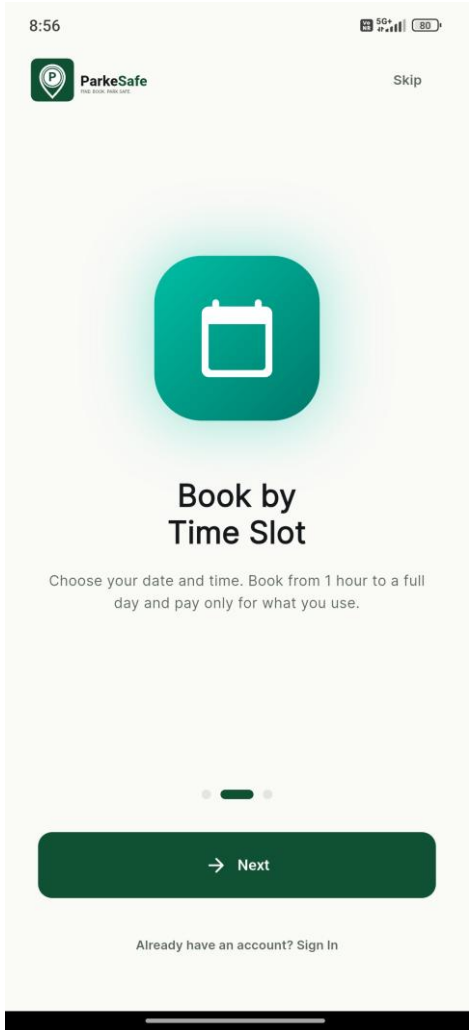
✓ Automated digital booking tickets and refund processing.

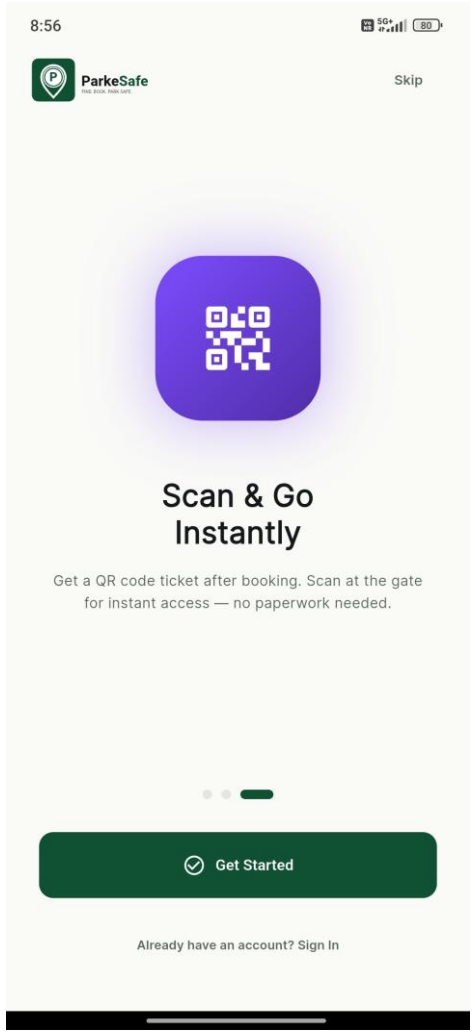
APPLICATION FLOW

Onboarding Screens

Interactive onboarding screens introducing users to nearby search, slot booking, and QR tickets.



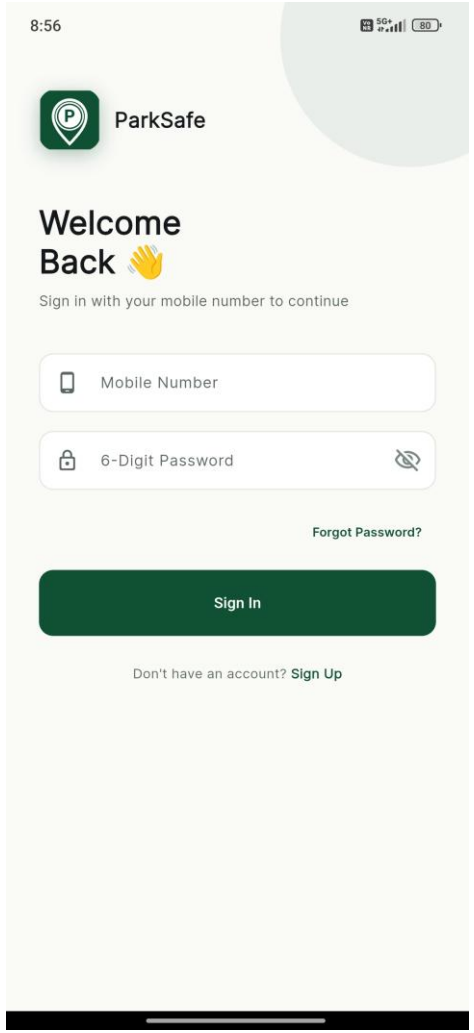


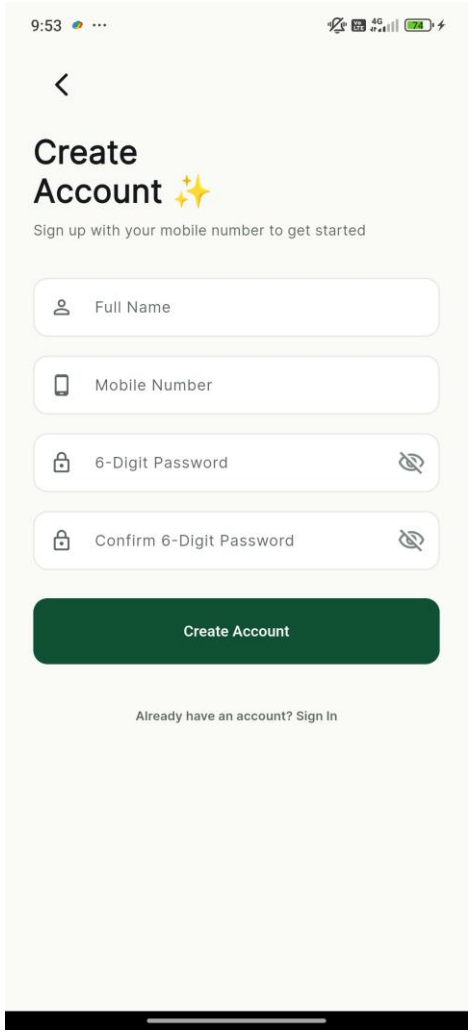


AUTHENTICATION

Login and Registration

Secure mobile authentication supporting user registration and returning user sign-in.

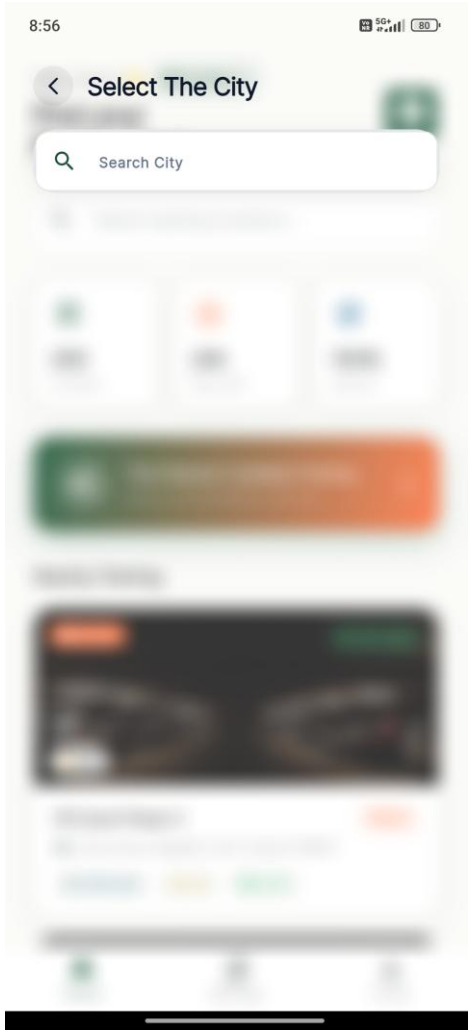


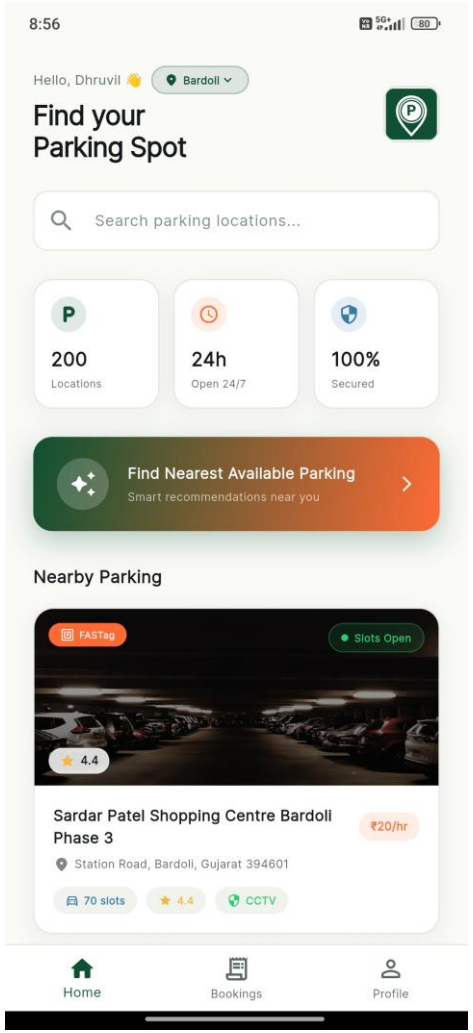


CUSTOMER INTERFACE

City Selection and Customer Home

Select preferred city and explore available parking locations with live search and statistics.

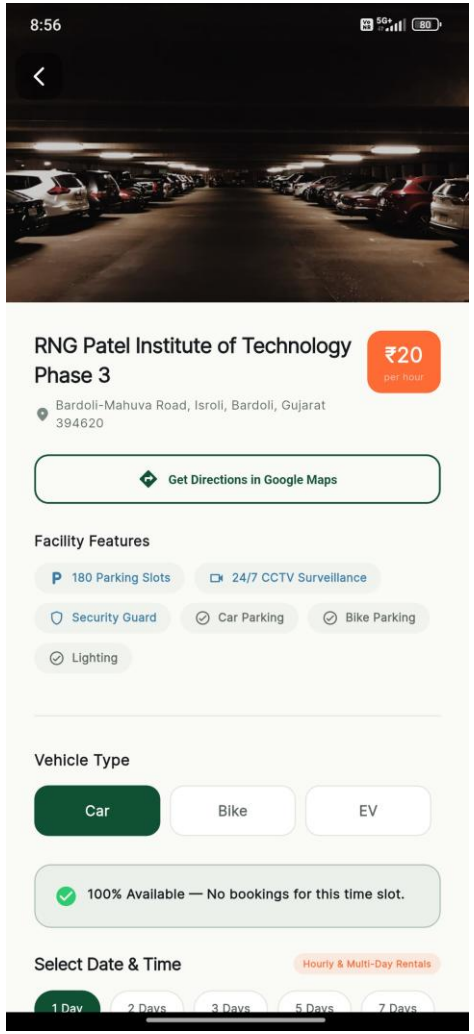


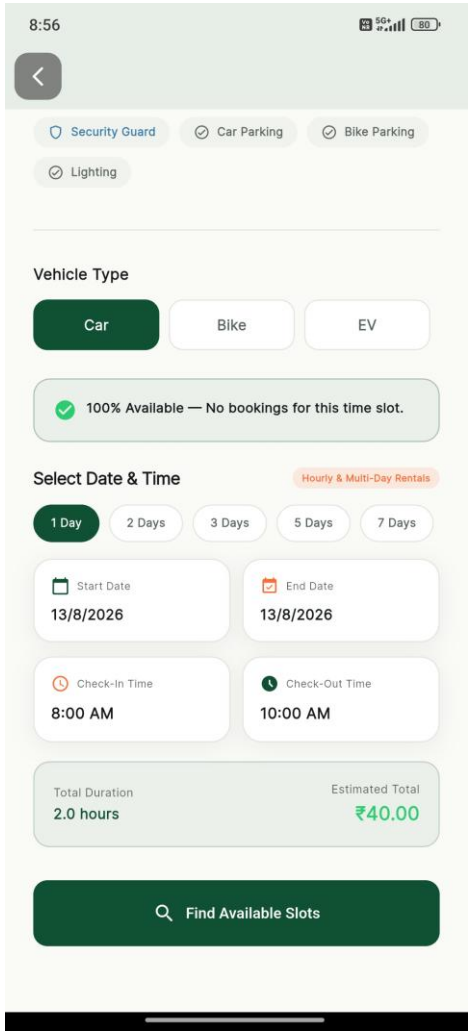


BOOKING PROCESS

Parking Details and Vehicle Selection

View location amenities, hourly pricing, Google Maps navigation, and vehicle type selection.

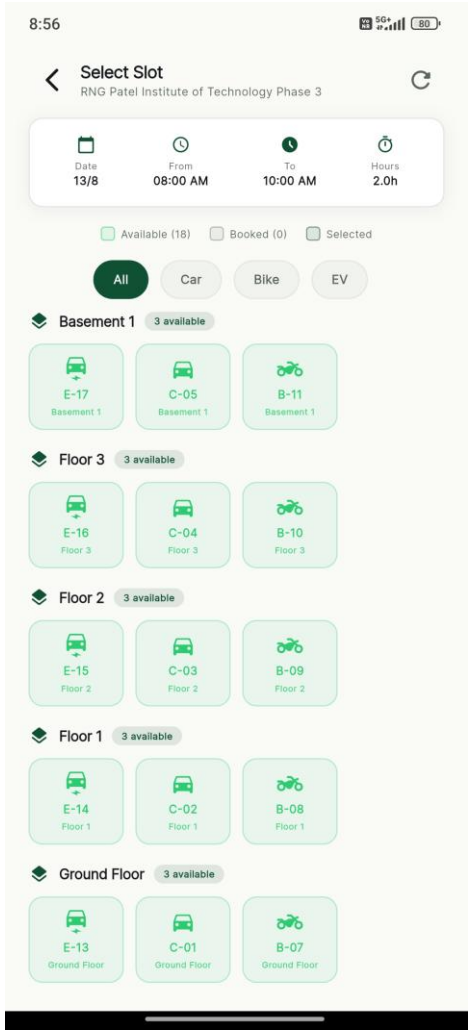


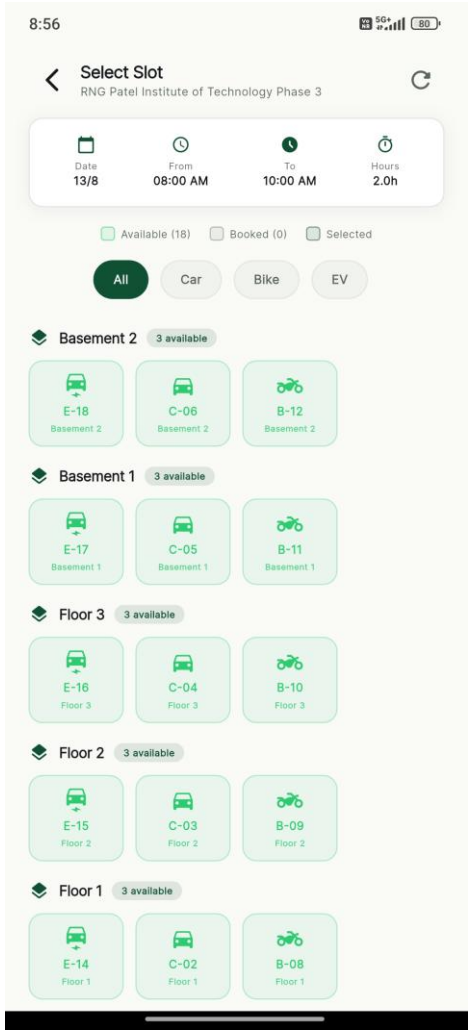


SLOT ALLOCATION

Floor-Wise Slot Selection

Browse different floors (Basement 1, Basement 2, Ground Floor) and select an available parking slot.

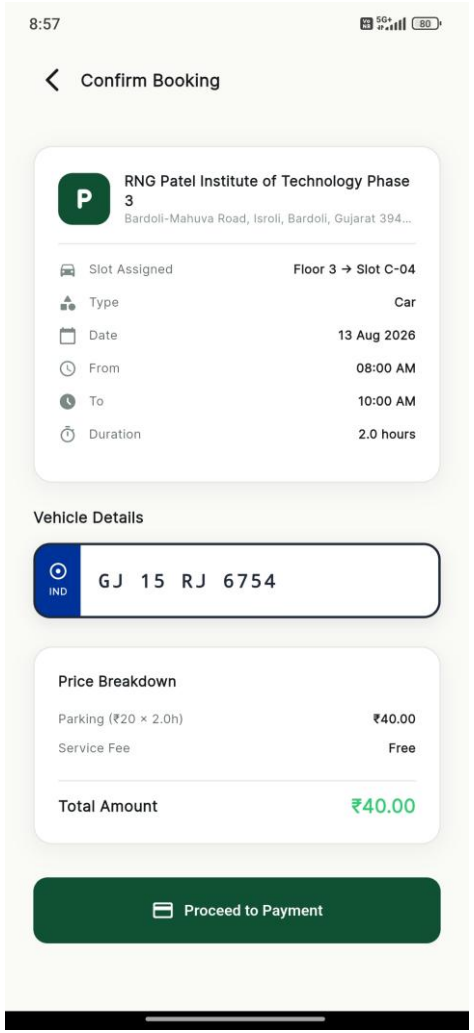


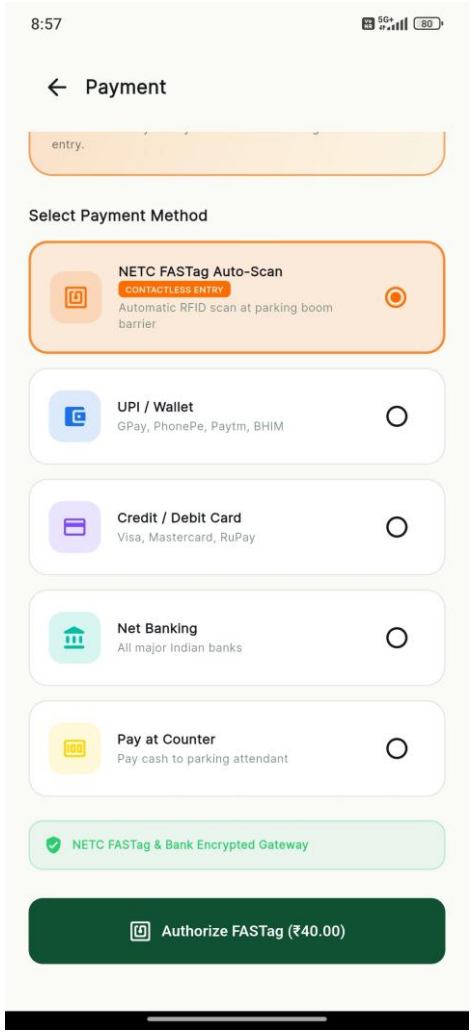


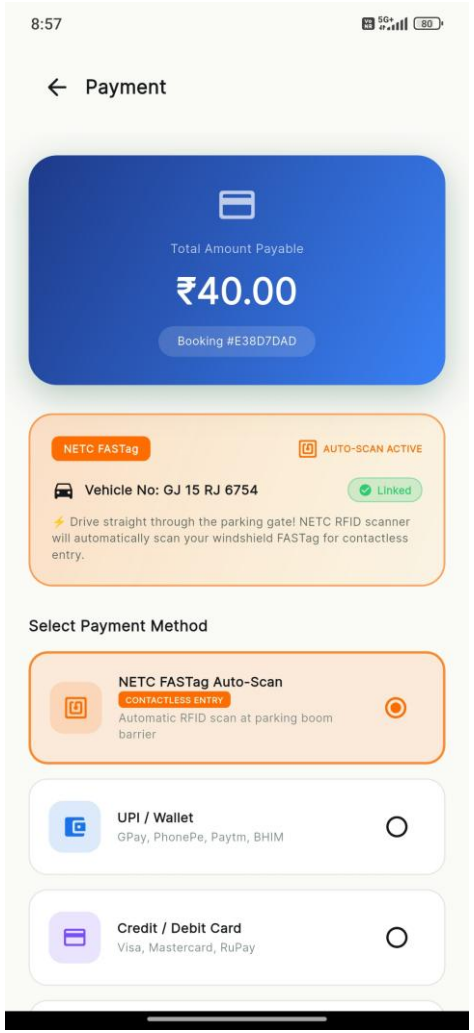
CHECKOUT FLOW

Booking Confirmation and Payment

Review reserved slot timing and choose payment via NETC FASTag Auto-Scan or UPI / Wallet.



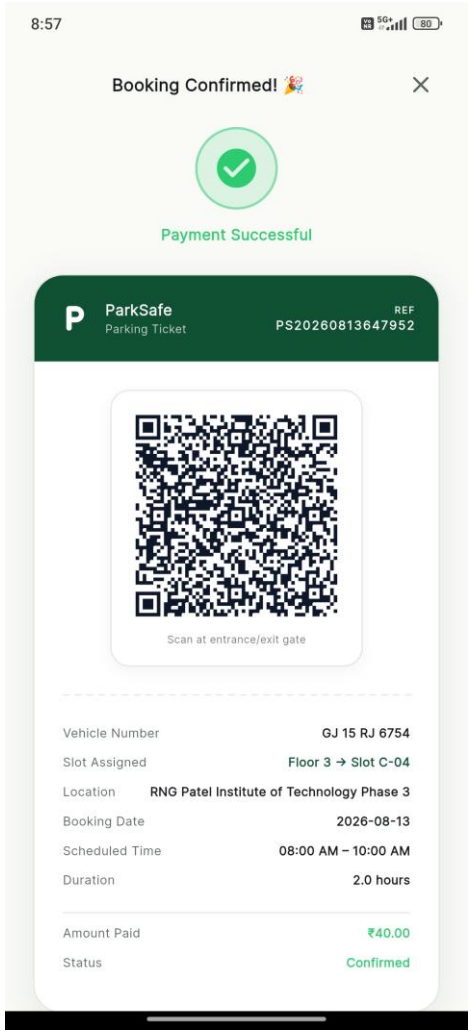


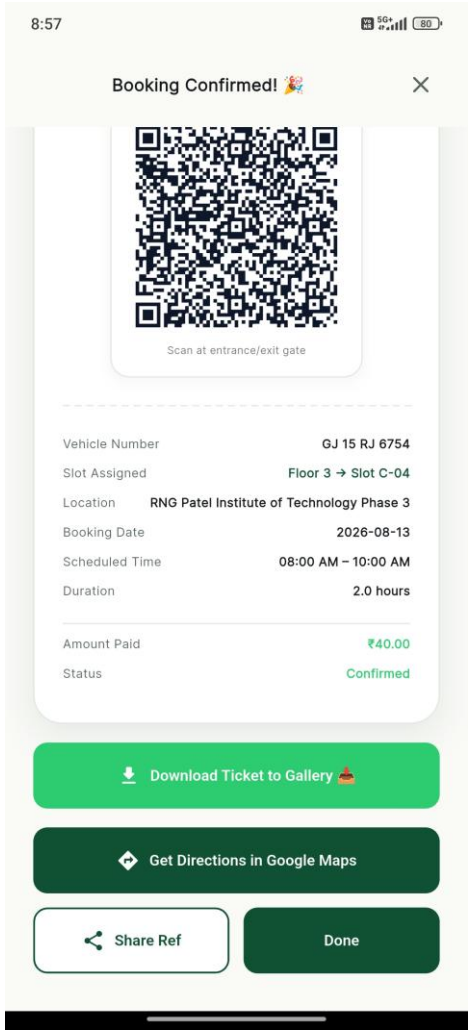


DIGITAL TICKET

Digital Booking Ticket & QR Code

Instant booking confirmation generating a digital QR pass containing slot and vehicle details.

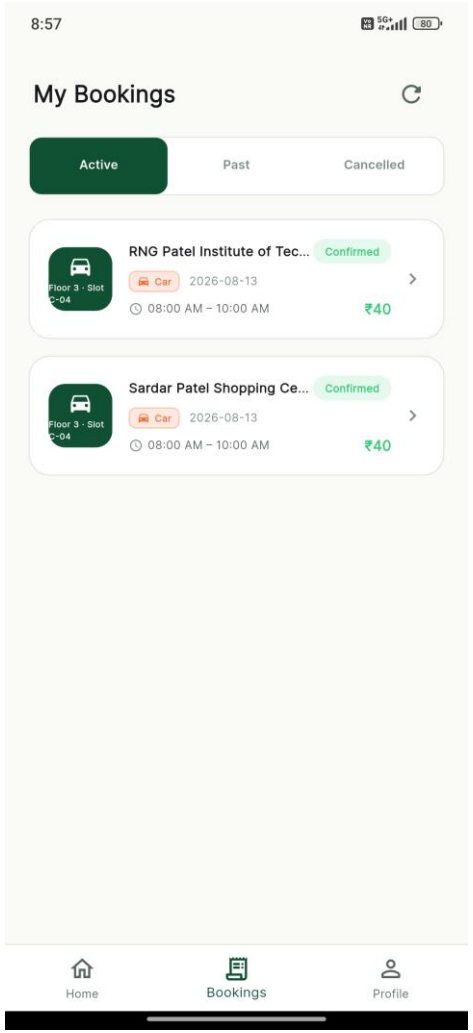


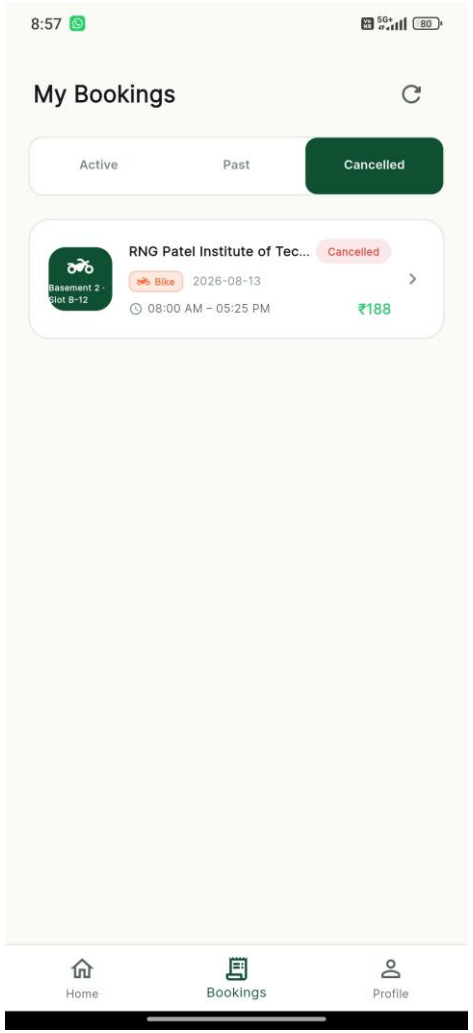


BOOKING MANAGEMENT

My Bookings (Active and Past)

Customers can track their active parking sessions and access past booking history.

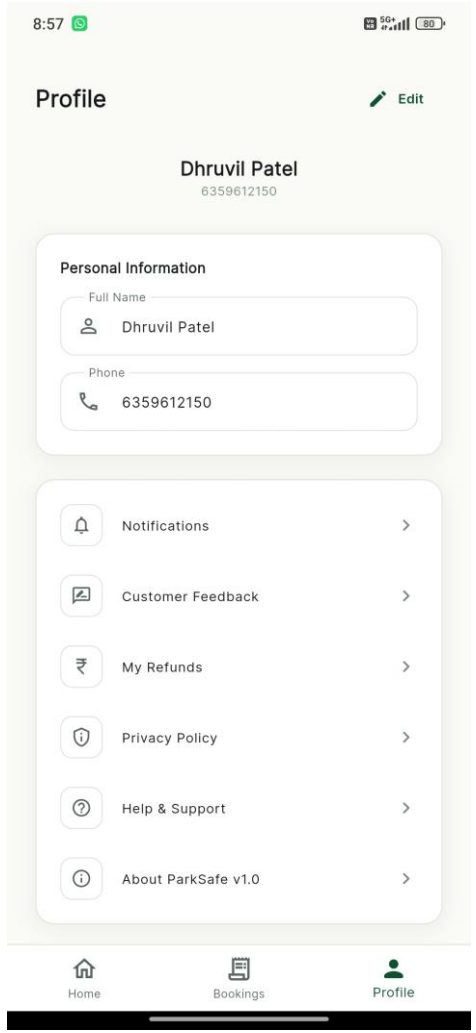




ACCOUNT SETTINGS

Customer Profile

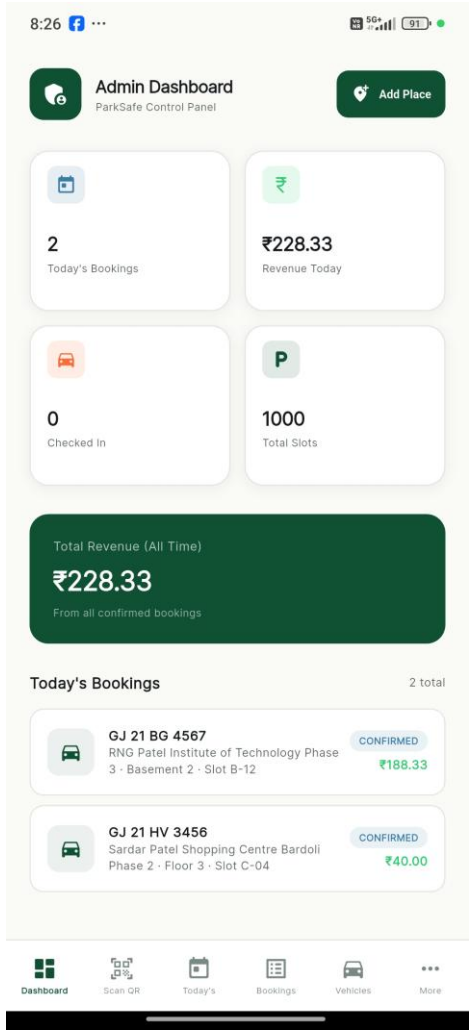
Customers manage their personal information, saved vehicles, and account preferences.

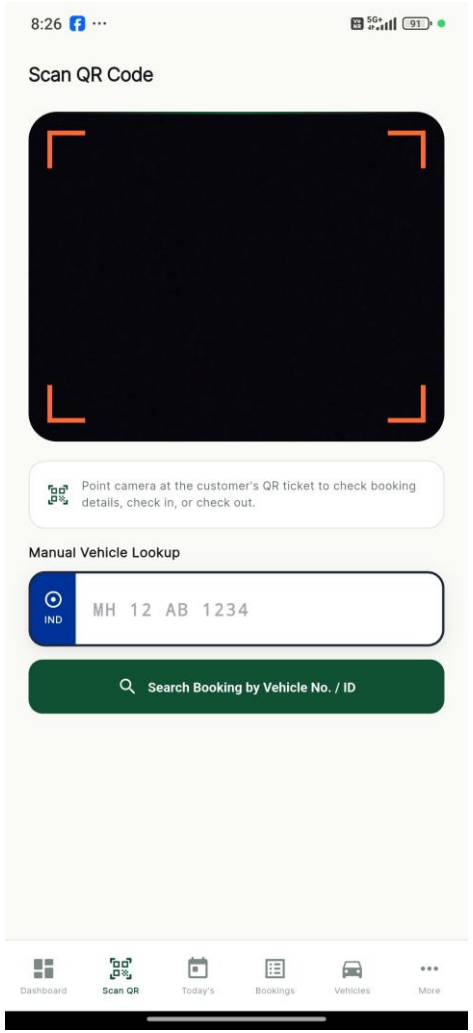


ADMIN OPERATIONS

Admin Dashboard and QR Scanner

Live occupancy and revenue dashboard paired with camera QR scanner for instant check-in.

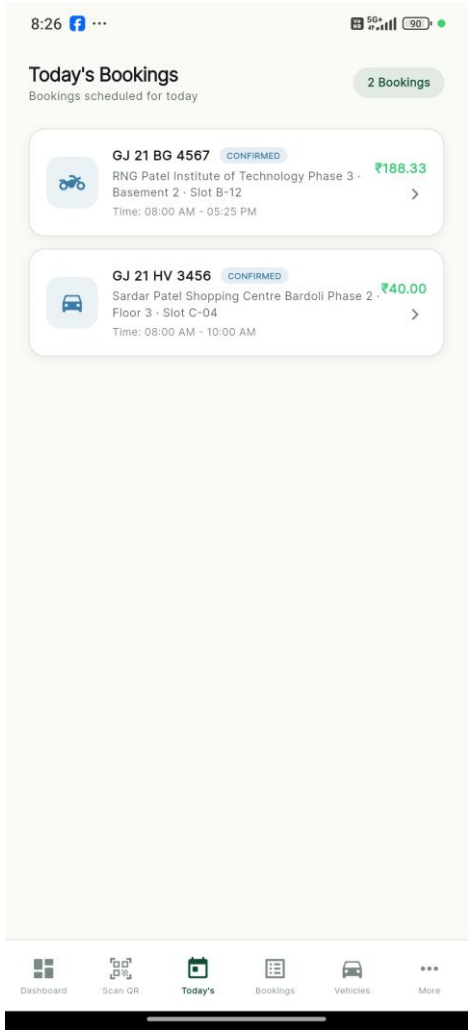


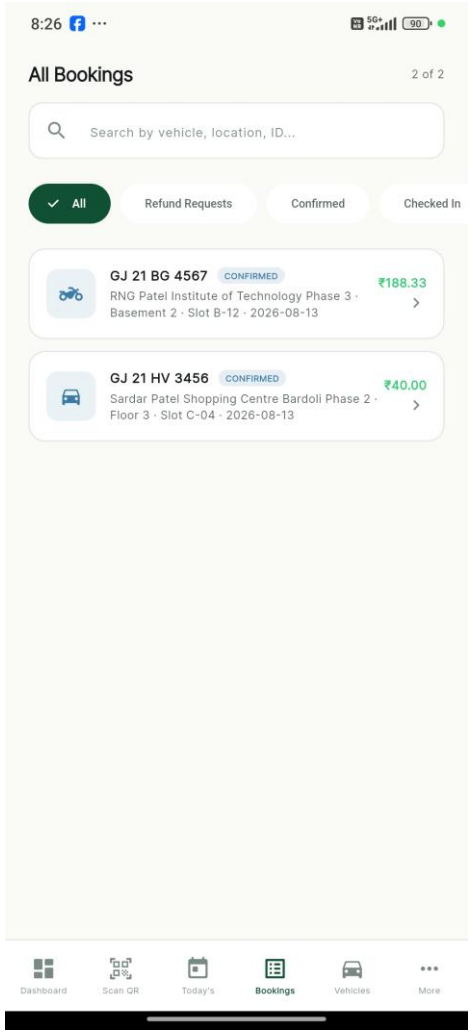


ADMIN OPERATIONS

Admin Booking Management

Real-time monitoring of today's scheduled arrivals and comprehensive all-bookings records.

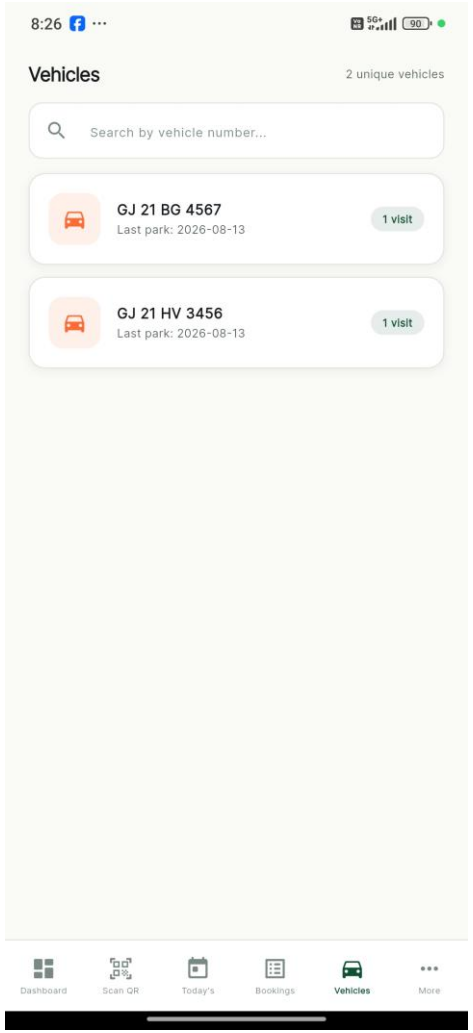




ADMIN OPERATIONS

Vehicles and Slot Management

Detailed parked vehicle logs and multi-floor slot distribution (Car, Bike, and EV spots).

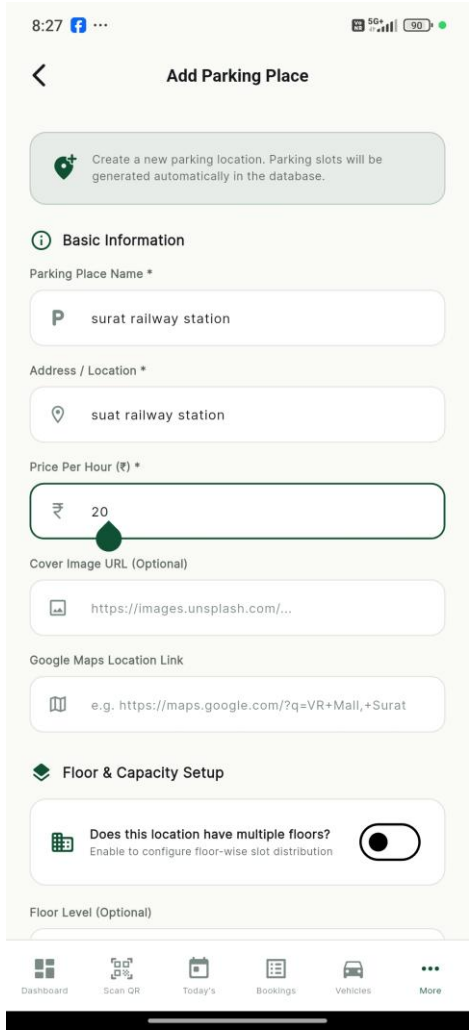


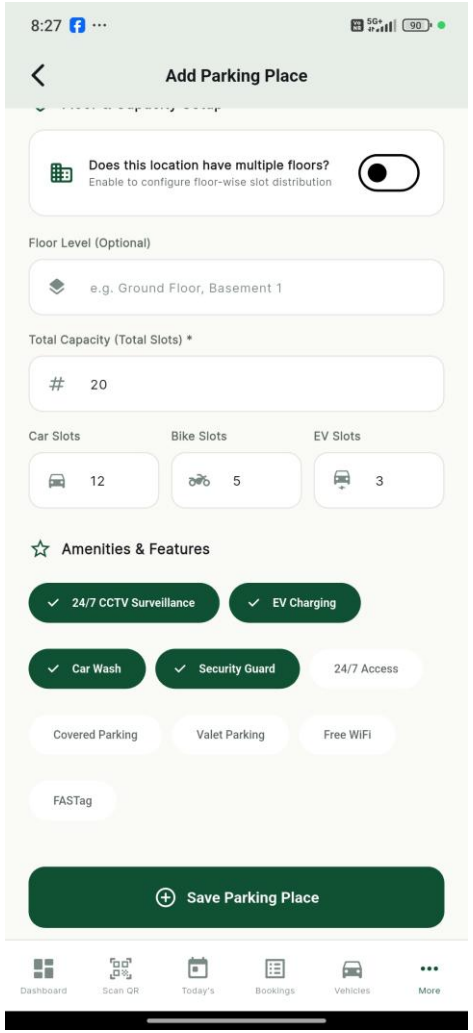


ADMIN OPERATIONS

Parking Location and Floor Configuration

Admins can add new parking locations and configure multi-floor slot distribution.

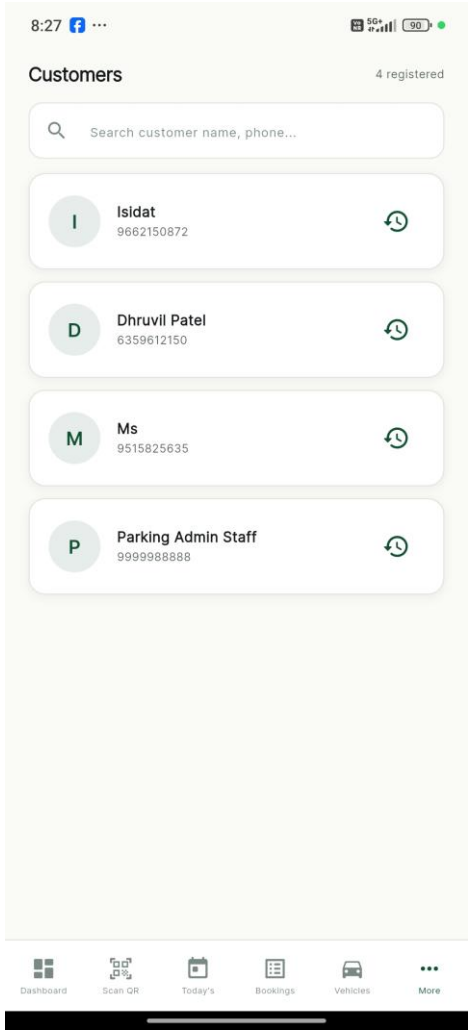


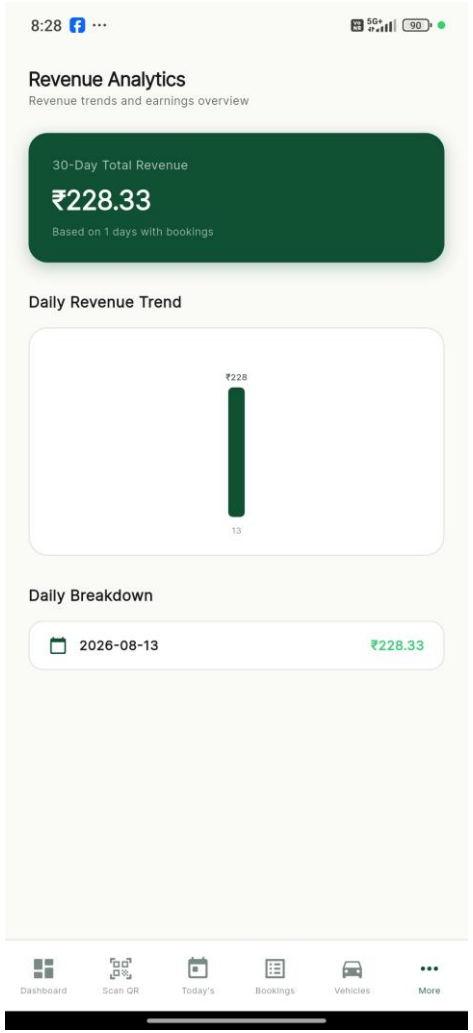


ADMIN OPERATIONS

Customer List and Revenue Analytics

Directory of registered users and visual 30-day parking revenue trend analytics.

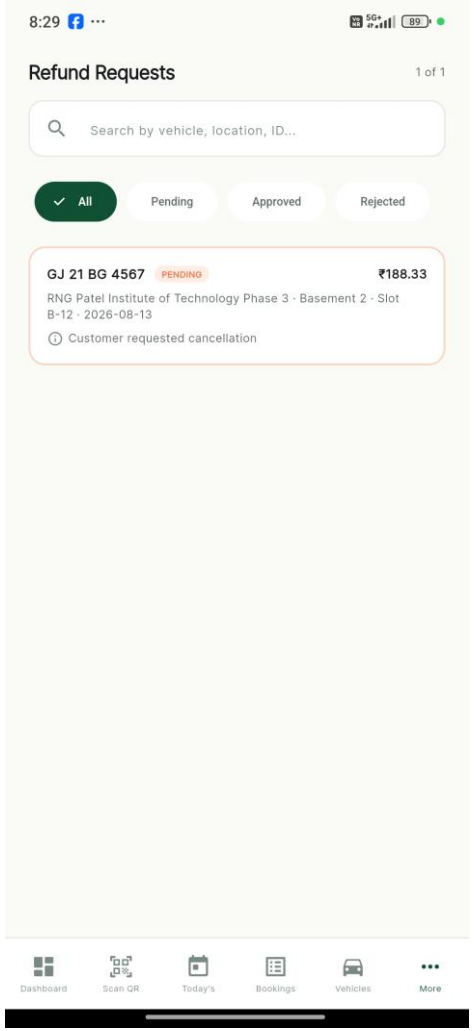




ADMIN OPERATIONS

Refund Management

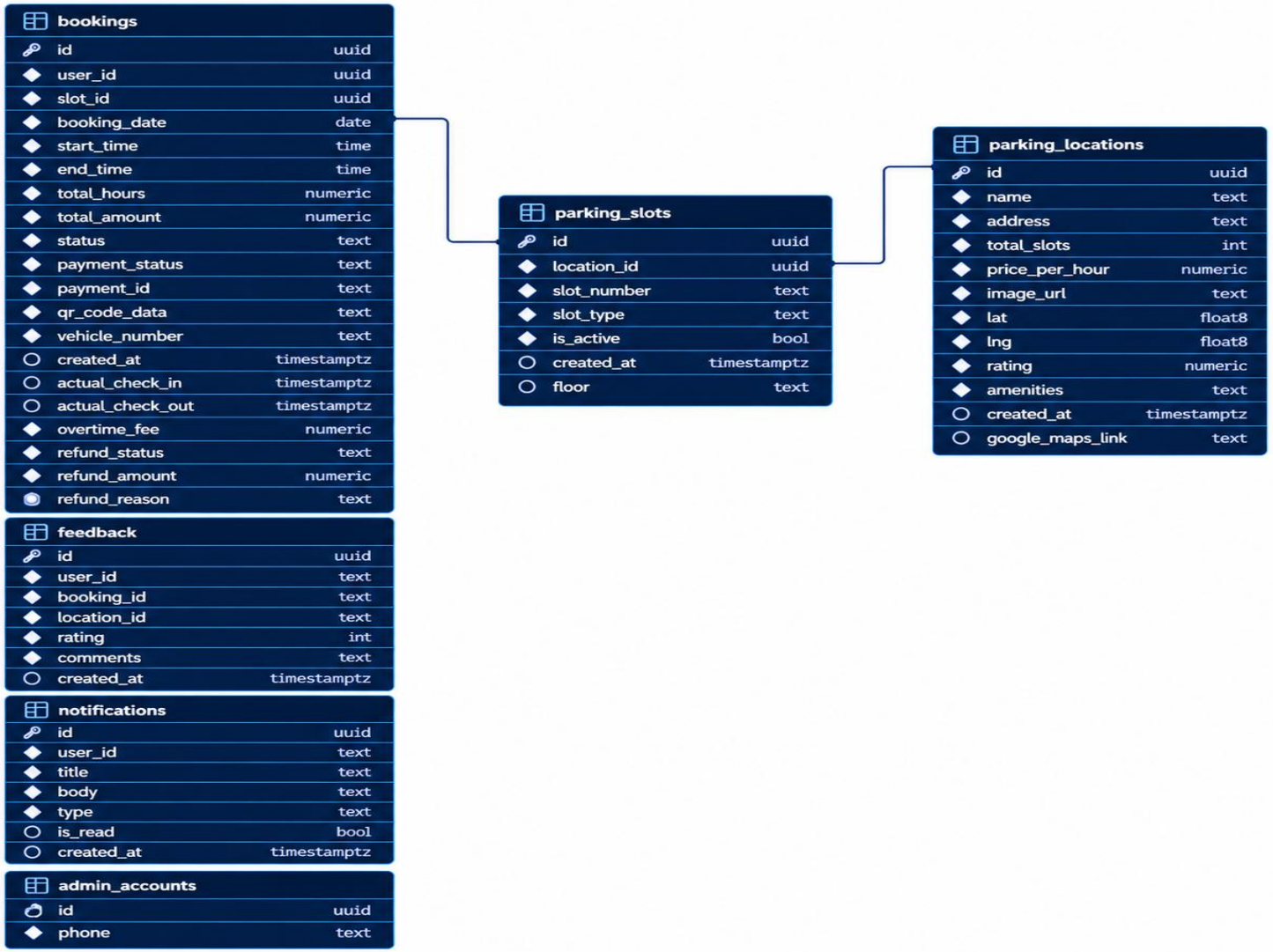
Admins review customer cancellation refund requests and approve or reject them directly.



BACKEND & DATABASE

Database Schema & Entity Architecture

Relational architecture in Supabase PostgreSQL: profiles, admin_accounts, parking_locations, slots, bookings, and feedback.



PROJECT TECHNOLOGIES

References & Technology Stack

Flutter & Dart

Cross-platform mobile and web application framework delivering high performance and responsive UI.

Supabase & PostgreSQL

Scalable cloud database backend, real-time data synchronization, and row-level security.

Provider State Management

Clean and reactive state management architecture connecting UI components to backend services.

GoRouter Navigation

Declarative and structured deep-linking routing for seamless multi-screen navigation flow.

Mobile Scanner & QR Engine

High-speed camera barcode/QR scanner integration for instant gate check-in and check-out.

Geolocator & Google Maps

Real-time location detection and GPS route navigation directing users to parking locations.

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Thank You

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